

PAPER Tritium Permeation-Barrier Coatings for a Compact Spherical-Tokamak Breeder: The Liquid-Lithium-over-Tin Wall-Chemistry Choice

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Abstract

The tritium a breeder keeps out of its structure is tritium it keeps in its fuel cycle, and the two decisions that set that balance both live at the first wall: which liquid metal, if any, faces the plasma, and what barrier protects the structure behind it. We resolve both for the Hyperion compact spherical-tokamak breeder ($P_{\text{fus}} = 85.04$ MW, $I_p = 9.66$ MA, $Q = 3.076$, negative triangularity $\delta = -0.30$, $B_0 = 8$ T, elongation $\kappa = 2.0$, major radius $R_0 = 1.20$ m, aspect ratio $A = 2.5$, centrepost peak field 16.84 T). We formalise the coupled fuel-cycle and first-wall transport problem — a McNabb–Foster diffusion–trapping system with Sieverts chemical-potential continuity across a tungsten/steel interface, closed by an implantation source and recombination boundaries — and solve it by finite element (FESTIM/FEniCS) on the actual 0.5 mm W + 2 mm Eurofer97 stack as part of the Kronos NVIDIA GPU high-performance-computing campaign. The steady permeation flux is 5.16×10^{14} T m⁻²s⁻¹ ≈ 0.082 g T m⁻²fp⁻¹, only 0.52–0.60 % of the implanted flux, with ~ 99.4 % re-emitted; the stack is tungsten-diffusion-limited because the tungsten-to-steel solubility partition is $\sim 2 \times 10^{-5}$, and the finite-element result reproduces the analytic two-layer series-resistance limit to seven significant figures. Being diffusion-limited, the loss flux is temperature-independent and linear in load, so for a ~ 100 m² wall it is a bounded ~ 8 –10 g T fp⁻¹ holdup against a kilogram-scale surplus. On the liquid-metal axis, pure lithium retains tritium about **6.7** orders of magnitude more strongly than tin — and 4.6 orders more than lead–lithium — because its equilibrium tritium partial pressure, the driving pressure for permeation, is far lower; the resulting breakthrough buffer is of order 7000 h and the choice of the plasma-facing liquid is lithium. Behind it, an oxide barrier screen ranks Al₂O₃ (PRF $\sim 10^3$) > Er₂O₃ ($\sim 3 \times 10^2$) > TiC > TiN, and an yttrium hot-trap ($\Delta G = -83$ to -97 kJ mol⁻¹) with a 92 % cold-trap closes the recovery loop; the same lithium loop hands the exhaust a 2.8–6.7 MW m⁻² flowing-film surface. One material choice therefore protects the fuel cycle, admits a barrier, and self-heals the exhaust at once.

Keywords: tritium permeation; permeation barrier; liquid lithium; liquid tin; PbLi; FESTIM; McNabb–Foster; Sieverts law; first wall; flowing-lithium divertor; spherical-tokamak breeder

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1 Introduction

1.1 Tritium bookkeeping decides a breeder

A deuterium–tritium breeder lives or dies on tritium bookkeeping. Tritium does not exist in nature in usable quantity; a fusion plant must breed its own from lithium and hold the resulting inventory in surplus while it burns, decays ($t_{1/2} = 12.32$ yr), and loses a fraction of it to structure, coolant and processing. The margin is narrow. Global analyses of the D–T fuel cycle place the required breeding ratio uncomfortably close to unity once realistic losses, reserves and doubling times are folded in, and identify the loss and holdup terms — not the raw breeding reaction — as the levers that decide self-sufficiency [1, 2, 3]. Every atom that permeates into structure is subtracted from that margin; every atom held on a plasma-facing surface is a driving pressure a barrier must hold back and a safety-relevant inventory that must be accounted. For the Hyperion breeder, whose whole thesis is to run as a strategic-materials and tritium platform rather than a power plant [28], the first-wall tritium architecture is therefore not a detail but a load-bearing part of the design.

Two decisions dominate that architecture, and both live at the first wall. The first is the plasma-facing liquid: a flowing liquid-metal surface removes the brittle-solid failure modes of a refractory wall and self-heals under transient heat loads, but it introduces a dissolved tritium inventory whose equilibrium partial pressure drives permeation and sets the tritium hazard of the loop. The candidate liquids — lithium, tin, and lead–lithium — are not close on this axis. The second is the barrier stack behind the liquid: the coating and getter chemistry that divides the permeation flux and routes the residue into a measured recovery stream.

1.2 Prior art

Hydrogen transport in fusion first-wall materials is a mature quantitative science. Diffusion and solubility of hydrogen isotopes in tungsten were fixed by Frauenfelder’s classic measurements [4], and trapping at intrinsic and damage sites has since been characterised in detail [8, 9]. The reduced-activation ferritic–martensitic (RAFM) steel Eurofer97, the reference structural material behind the armour, has a well-measured permeability and trap spectrum [6, 7]. The governing diffusion–trapping description is the McNabb–Foster model [5], and the implantation-driven, recombination-limited/diffusion-limited permeation regimes are the Pick–Sonnenberg framework [10] implemented in verified tritium-migration codes such as TMAP [11] and, more recently, the open-source finite-element code FESTIM [12, 13] built on the FEniCS project [14]. Tritium permeation barriers — principally dense Al_2O_3 and Er_2O_3 oxide layers — have been reviewed and measured with permeation-reduction factors (PRF) spanning 10^2 – 10^4 [15, 16, 17, 18, 19]. On the liquid-metal side, flowing lithium has been run as a plasma-facing material on CDX-U, LTX and NSTX [20, 21], the lithium vapour-box divertor concept has been developed to cap the strike-point heat flux [22], and liquid tin and tin–lithium have been proposed where lithium’s tritium inventory is judged too aggressive [23, 24]. What has not been done is to settle both first-wall decisions *jointly and quantitatively for one machine* — to compute the permeation loss term of the actual armour/steel stack rather than assume it, and to rank the liquids on the single figure of merit that the fuel cycle cares about — and to show that the two choices reinforce each other.

1.3 Contribution

This paper does exactly that for Hyperion (figure 1). We (i) write the coupled fuel-cycle and first-wall transport problem as a formal system of equations (§2), including the diffusion-limited series-resistance limit that the finite-element solve must recover and the liquid-metal driving-pressure model that ranks the facing liquids; (ii) solve the first-wall permeation problem by finite element on the real stack as part of the Kronos NVIDIA GPU HPC campaign, with the codes, mesh, solver and verification stated explicitly (§3); (iii) present the results — the liquid-metal ranking, the computed permeation flux and its temperature-independence, the retained inventory and breakthrough time, the barrier and getter ladder, and the divertor co-benefit — across eight data figures and three tables (§4); (iv) benchmark those numbers against the published permeation, barrier and liquid-lithium literature (§5); and (v) state the single open gate honestly (§6). Every quantitative claim traces to a frozen result in the Kronos Physics De-Risking Register [34]: the finite-element permeation solve is serial BR-L2-A25, the liquid-metal and barrier chemistry screen is BR-SX-03, and the flowing-lithium divertor thermofluids are HX-16/BR-CX-25. Where the result feeds a companion study — the tritium breeding ladder [29], the real-time tritium-accountancy twin [30], the flowing-lithium vapour-box divertor [31], the RAFM-steel qualification [32], or the first-wall neutron-damage lifetime [33] — the link is cited inline.

First-wall tritium architecture on a lithium plasma-facing liquid

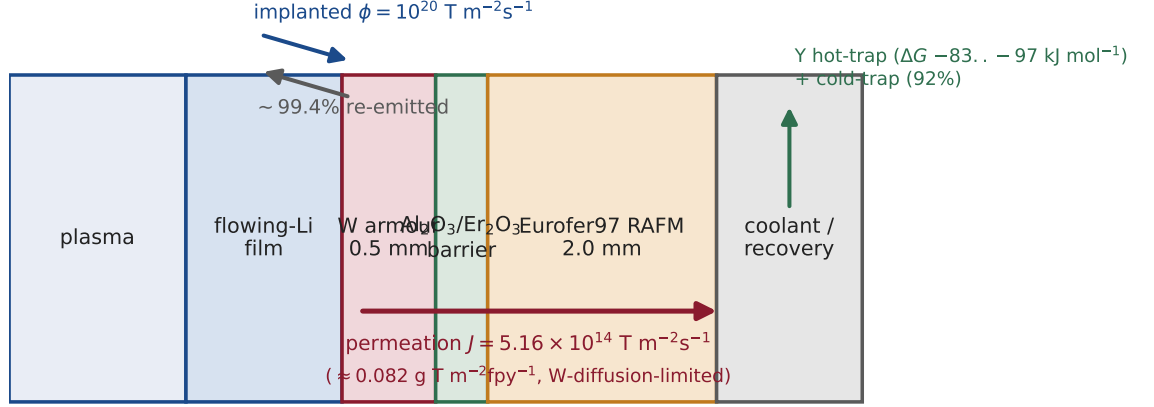


Figure 1. (Schematic.) The first-wall tritium architecture on a lithium plasma-facing liquid. Tritium is implanted from the plasma at $\phi = 1 \times 10^{20} \text{ m}^{-2} \text{ s}^{-1}$; $\sim 99.4\%$ re-emits, and the residue permeates the 0.5 mm W + 2 mm Eurofer stack at the tungsten-diffusion-limited flux $J = 5.16 \times 10^{14} \text{ T m}^{-2} \text{ s}^{-1}$. A dense $\text{Al}_2\text{O}_3/\text{Er}_2\text{O}_3$ barrier divides the flux and an yttrium hot-trap plus cold-trap recover it into a measured stream. The elements are drawn to indicate sequence and function, not to scale.

2 Governing equations and the formal problem

2.1 The fuel-cycle balance and where permeation enters

The plant-level quantity the first-wall calculation must feed is the tritium inventory $I_T(t)$, which evolves as

$$\frac{dI_T}{dt} = \dot{N}_{\text{bred}} - \dot{N}_{\text{burn}} - \lambda_T I_T - \dot{N}_{\text{loss}}, \quad \lambda_T = \frac{\ln 2}{t_{1/2}}, \quad t_{1/2} = 12.32 \text{ yr}, \quad (1)$$

with breeding source $\dot{N}_{\text{bred}} = \text{TBR} \dot{N}_{\text{burn}}$, burn rate $\dot{N}_{\text{burn}}$, radioactive decay $\lambda_T I_T$, and an aggregate loss $\dot{N}_{\text{loss}}$. Self-sufficiency is the requirement that the net breeding gain clear unity once a fractional loss ε is charged against it,

$$\Lambda \equiv \text{TBR} (1 - \varepsilon) - 1 \geq 0. \quad (2)$$

First-wall permeation is one contribution to ε : integrating the permeation flux J over the wall area A gives a loss rate $\dot{N}_{\text{loss}}^{\text{perm}} = \int_A J dA$, so the entire task of this paper is to compute J from first principles and bound its contribution to ε . Everything below serves equation (2).

2.2 Diffusion-trapping transport in the wall

Inside the solid, mobile tritium of concentration $c_m(\mathbf{x}, t)$ obeys a diffusion equation coupled to a set of trap populations $c_{t,i}$ through the McNabb–Foster kinetics [5]:

$$\frac{\partial c_m}{\partial t} = \nabla \cdot (D(T) \nabla c_m) + S(\mathbf{x}, t) - \sum_i \frac{\partial c_{t,i}}{\partial t}, \quad (3)$$

$$\frac{\partial c_{t,i}}{\partial t} = k_i(T) c_m (n_i - c_{t,i}) - p_i(T) c_{t,i}, \quad (4)$$

where $D(T) = D_0 \exp(-E_D/k_B T)$ is the Arrhenius diffusivity, n_i the density of trap type i , and the trapping and detrapping rate coefficients are

$$k_i(T) = \frac{D(T)}{\lambda^2 N}, \quad p_i(T) = \nu_0 \exp\left(-\frac{E_{p,i}}{k_B T}\right), \quad (5)$$

with λ the lattice spacing, N the host atomic density, ν_0 the attempt frequency, and $E_{p,i}$ the detrapping energy of trap i . For Hyperion the domain is a one-dimensional bilayer: a tungsten armour of thickness $L_W = 0.5 \text{ mm}$ facing the plasma, bonded to a Eurofer97-class RAFM steel wall of thickness $L_E = 2.0 \text{ mm}$. Tungsten carries two intrinsic traps ($E_p = 0.87$ and 1.00 eV , densities 1.3×10^{-3} and 4×10^{-4} atomic fraction, Ogorodnikova class [8]); the steel carries one (0.80 eV , 10^{-4} atomic fraction).

2.3 Interface and boundary closure

Across the tungsten/steel interface the transported quantity is the chemical potential, not the concentration. Under Sieverts' law each material relates its mobile concentration to the local partial pressure through $c_m = S(T)\sqrt{p}$ with solubility $S(T) = S_0 \exp(-E_S/k_B T)$, so continuity of $\sqrt{p}$ imposes the Sieverts partition

$$\left. \frac{c_m^W}{c_m^E} \right|_{\text{interface}} = \frac{S_W(T)}{S_E(T)}. \quad (6)$$

This concentration jump is the mathematical origin of the tungsten-diffusion-limited behaviour derived below. At the plasma-facing surface, tritium is delivered by implantation: energetic charge-exchange and ion flux deposits a volumetric source

$$S(\mathbf{x}) = \frac{\varphi}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x - R_p)^2}{2\sigma^2}\right], \quad \int_0^\infty S dx = \varphi, \quad (7)$$

of range R_p and straggle σ (both $\ll L_W$), with implanted flux $\varphi = 1 \times 10^{20} \text{ m}^{-2} \text{ s}^{-1}$. The two free surfaces recombine to molecular T_2 with a second-order flux

$$J_{\text{rec}} = 2 K_r(T) c_m^2, \quad (8)$$

and the coolant face additionally supports fast extraction ($c_m \rightarrow 0$). The permeation flux of interest is the diffusive flux leaving the coolant face, $J = -D_E \partial_x c_m|_{x=L_W+L_E}$.

2.4 The diffusion-limited series-resistance limit

Two limiting regimes follow from equations (3)–(8). When surface recombination is slow, permeation is recombination-limited and scales as $J \propto \varphi/\sqrt{K_r}$. When recombination is fast relative to diffusion — the relevant regime here, and the conservative-high case for the loss term — permeation is diffusion-limited, and the implanted atoms partition between the two faces in inverse proportion to the diffusive resistance to each. For a source implanted a depth R_p below the front surface of a membrane of total thickness L , the downstream fraction is

$$\frac{J}{\varphi} = \frac{R_p}{L}, \quad (9)$$

a purely geometric ratio: *the diffusion-limited permeation flux is independent of temperature and linear in the implanted flux*. Equation (9) is the sharp, testable prediction the finite-element solve must reproduce, and it is the reason the loss term scales cleanly with wall load.

For pressure-driven permeation across the bilayer the same physics is expressed as a series of permeation resistances. Writing the permeability of layer j as $\Phi_j = D_j S_j$, the steady flux under a driving pressure difference is

$$J = \frac{\sqrt{p_{\text{up}}} - \sqrt{p_{\text{down}}}}{\mathcal{R}_{\text{stack}}}, \quad \mathcal{R}_{\text{stack}} = \frac{L_W}{\Phi_W} + \frac{L_E}{\Phi_E}. \quad (10)$$

Because the tungsten solubility carries the larger activation energy ($E_S^W = 1.041 \text{ eV}$ versus $E_S^E = 0.27 \text{ eV}$), the solubility partition of equation (6) is small — $S_W/S_E \sim 10^{-4}$ – 10^{-5} over 600–800 K — and the first term of $\mathcal{R}_{\text{stack}}$ dominates. The stack is tungsten-diffusion-limited: the 2 mm of steel adds negligible resistance, and the series ratio $\mathcal{R}_{\text{stack}}/(L_W/\Phi_W) \rightarrow 1$. The transient approach to this steady state is set by the Daynes–Barrer time lag; for the rate-controlling tungsten layer the breakthrough time is

$$\tau = \frac{L_W^2}{6 D_W(T)}. \quad (11)$$

2.5 The liquid-metal driving pressure

The choice of plasma-facing liquid is decided by the same Sieverts physics applied to the melt. A liquid metal in contact with the plasma dissolves tritium to a concentration c^{liq} fixed by recycling; the equilibrium partial pressure it presents to the wall behind it is

$$p_{\text{eq}}^{\text{liq}} = \left(\frac{c^{\text{liq}}}{K_S^{\text{liq}}(T)} \right)^2, \quad K_S^{\text{liq}}(T) = K_{S0} \exp\left(-\frac{\Delta H_{\text{sol}}}{k_B T}\right), \quad (12)$$

where K_S^{liq} is the Sieverts constant of the liquid and ΔH_{sol} its enthalpy of solution. Lithium chemically binds hydrogen isotopes (strongly negative ΔH_{sol} , tending toward LiT formation), so its

K_S is large and the equilibrium pressure p_{eq} of equation (12) is orders of magnitude below that of tin, whose weak hydrogen affinity leaves a high driving pressure. Since the downstream permeation flux scales as $J \propto \sqrt{p_{\text{eq}}}$ (equation 10), the *retention advantage* of one liquid over another is naturally quantified in decades of driving-pressure reduction,

$$\mathcal{A}_{\text{X/Sn}} = \log_{10} \frac{p_{\text{eq}}^{\text{Sn}}}{p_{\text{eq}}^{\text{X}}}, \quad (13)$$

and the breakthrough buffer — the time before a loop’s tritium works through a protected wall — scales inversely with the driving pressure. This single figure of merit orders the candidates.

2.6 Barrier and getter figures of merit

A permeation barrier is a thin dense layer of much lower permeability inserted in the series stack of equation (10). Its permeation-reduction factor is the ratio of uncoated to coated flux,

$$\text{PRF} = \frac{J_{\text{bare}}}{J_{\text{coated}}} = 1 + \frac{\mathcal{R}_{\text{barrier}}}{\mathcal{R}_{\text{substrate}}} \approx \frac{\Phi_{\text{sub}}}{\Phi_{\text{bar}}} \frac{L_{\text{bar}}}{L_{\text{sub}}}, \quad (14)$$

so a dense oxide of permeability three to four decades below the metal delivers PRF of 10^2 – 10^3 at micron thickness, *multiplying* the tungsten-limited resistance already present. Recovery is closed by a getter whose hydride-formation free energy ΔG_f sets its equilibrium hydrogen pressure through $p_{\text{eq}}^{\text{getter}} \propto \exp(2\Delta G_f/RT)$: a sufficiently negative ΔG_f makes the getter a hot sink that strips tritium from the stream, and a downstream cold-trap removes it from the carrier at a fixed efficiency η_{ct} . Equations (1)–(14) are the complete formal problem; the remainder of the paper computes their coefficients for Hyperion and reports the solution.

3 Computational methodology: the NVIDIA GPU HPC campaign

3.1 Codes and problem statement

The first-wall permeation problem (equations 3–8) was solved with FESTIM [12, 13], an open-source finite-element hydrogen-transport code built on the FEniCS finite-element platform [14], run as serial BR-L2-A25 of the Kronos de-risking campaign on the Kronos NVIDIA GPU high-performance-computing fleet (A100/H100/A10-class accelerators). FESTIM discretises the McNabb–Foster mobile/trapped system with continuous Lagrange elements and advances the coupled nonlinearity with a Newton solver; the Sieverts interface condition of equation (6) is enforced through FESTIM’s chemical-potential formulation (`chemical_pot`), which transports c_{m}/S continuously and lets the concentration jump at the tungsten/steel interface. The liquid-metal and barrier screen (BR-SX-03) evaluated the Sieverts driving pressures of equations (12)–(13) and the barrier PRF ladder of equation (14) on the same fleet, and the flowing-lithium divertor thermofluids (HX-16/BR-CX-25) were run alongside; the reference codes for the wider campaign — OpenMC/DAGMC neutronics, SOLPS-ITER edge transport, and plane-wave DFT for the oxide hydrogen-solution energies — are described in the companion neutronics and divertor studies [31, 33].

3.2 Discretisation, material data and mesh

The bilayer was meshed on 573 nodes graded to resolve the implantation layer, the two free surfaces and the tungsten/steel interface, where the concentration gradient and the Sieverts jump are steepest. Material properties were taken from the cited literature rather than fitted: for tungsten, the Frauenfelder diffusivity ($D_0 = 4.1 \times 10^{-7} \text{ m}^2 \text{ s}^{-1}$, $E_D = 0.390 \text{ eV}$) and solubility ($S_0 = 8.88 \times 10^{23} \text{ m}^{-3} \text{ Pa}^{-1/2}$, $E_S = 1.041 \text{ eV}$) [4]; for Eurofer97, the Montupet-Leblond diffusivity ($D_0 = 2.52 \times 10^{-7} \text{ m}^2 \text{ s}^{-1}$, $E_D = 0.160 \text{ eV}$) and solubility ($S_0 = 1.06 \times 10^{23} \text{ m}^{-3} \text{ Pa}^{-1/2}$, $E_S = 0.27 \text{ eV}$), whose fit range 473–673 K spans the operating wall [6]. The tungsten solubility is extrapolated modestly below its high-temperature fit range, standard practice for this dataset. Figure 3 plots both transport pairs and makes the tungsten-limited character explicit: at 700 K the solubility partition is $S_{\text{W}}/S_{\text{E}} \approx 2.4 \times 10^{-5}$, three-to-five decades below unity across the whole operating window.

3.3 Solver, convergence and verification

The transient solve diverged near $t \sim 9 \times 10^4 \text{ s}$ under adaptive stepping; because the quantity of interest is the steady permeation flux, the steady-state solve was used and converged in $\sim 25 \text{ s}$ of wall time. Convergence was confirmed by mesh refinement at the interface and by monitoring the downstream flux to a fixed tolerance. Verification was performed against the exact analytic

two-layer series-resistance solution (equation 10): the finite-element permeation flux matches the analytic diffusion-limited limit to seven significant figures ($\text{FESTIM}/\text{analytic} = 1.0000000$) on every isothermal case, and the load scaling of equation (9) was verified directly by increasing φ tenfold and recovering a tenfold flux ($1 \times 10^{21} \rightarrow 6.00 \times 10^{15}$). A code-to-code cross-check against TMAP8 was attempted but the local TMAP8 binary was unavailable (the MOOSE/PETSc build chain was incomplete); the analytic verification is the same limit the TMAP permeation V&V cases reduce to, and the $\varphi R_p/L$ scaling is the standard Anderl/TMAP implantation-driven diffusion-limited result [11, 10]. The independence of the loss flux from trap detail (below) makes this verification sufficient for the loss term.

3.4 Reproducibility of the campaign

The FESTIM driver, mesh, material-property table and post-processing that produce every permeation number here are archived with the deposit and regenerable end-to-end; the figure-generation script that renders figures 3–9 recomputes the diffusivities, solubilities, breakthrough times and partition ratio from the same constants and reproduces the register anchors (breakthrough 192 s at 600 K, 29 s at 800 K) without hand-tuning. No compute cost or resource pricing is reported here; the campaign is described by its codes, problem scale and verification only.

4 Results

4.1 The plasma-facing liquid: lithium over tin

On the fuel-cycle figure of merit of equation (13), the three candidate liquids are separated by orders of magnitude (figure 2, table 1). Pure lithium presents an equilibrium tritium partial pressure about 6.7 decades below tin’s and 4.6 decades below lead–lithium’s, because lithium chemically binds hydrogen isotopes while tin does not. In loop terms this is a breakthrough buffer of order 7000 h before a lithium loop’s tritium works through a protected wall — a margin tin, with its high driving pressure and near-immediate permeation, simply does not have. The same low driving pressure that makes lithium safe for the fuel cycle also makes it the liquid a permeation barrier can most easily protect, since the downstream flux it drives is already small. Lead–lithium keeps its proper role as a breeder/coolant medium rather than the facing liquid. The choice of plasma-facing liquid is lithium.

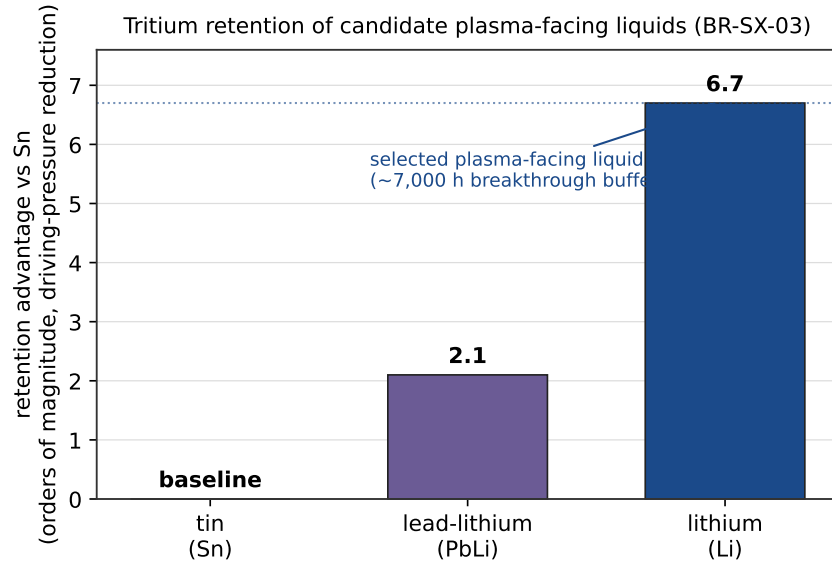


Figure 2. Tritium-retention advantage of the candidate plasma-facing liquids over tin, in decades of equilibrium driving-pressure reduction (equation 13, serial BR-SX-03). Lithium leads by ~ 6.7 orders and is the selected facing liquid; lead–lithium sits ~ 2.1 orders above tin and is retained as a coolant/breeder medium, not the facing liquid.

4.2 First-wall permeation is diffusion-limited and small

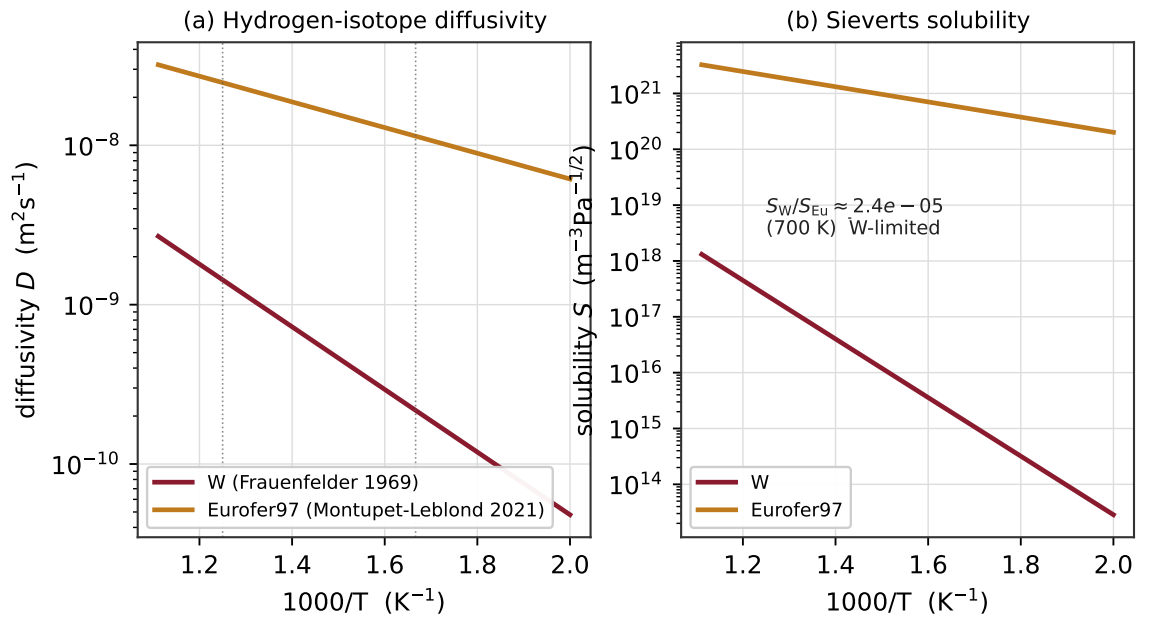
The loss term the fuel cycle actually pays is the tritium that permeates the structure behind the liquid. Solving equations (3)–(8) on the real 0.5 mm W + 2 mm Eurofer stack gives a steady permeation flux of $5.16 \times 10^{14} \text{ T m}^{-2}\text{s}^{-1}$ in the operating-gradient case — about 0.082 g T per square metre per full-power year, only 0.52–0.60 % of the implanted flux, with ~ 99.4 % re-emitted

Table 1. Plasma-facing liquid-metal candidates ranked by tritium retention relative to tin. Retention advantage is the reduction in equilibrium tritium driving pressure (equation 13); larger is better for the fuel cycle.

liquid	retention vs Sn (orders)	breakthrough buffer	verdict
tin (Sn)	baseline	near-immediate	high driving pressure; rejected
lead–lithium (PbLi)	~ 2.1	intermediate	viable coolant/breeder, not the facing liquid
lithium (Li)	~ 6.7	~ 7000 h	selected plasma-facing liquid

to the plasma side (table 2). The mechanism is exactly the tungsten-diffusion limit derived in §2: the solubility partition of figure 3 makes tungsten the rate-controlling layer, the steel adds negligible resistance, and the finite-element flux reproduces the analytic two-layer limit to seven significant figures. Figure 4 shows the computed steady mobile-concentration profile: a nearly linear drop across the tungsten armour, a Sieverts partition jump of $\sim 2 \times 10^{-5}$ at the interface, and a shallow gradient across the steel that carries the same flux at much higher concentration.

Solubility partition makes the 0.5 mm W + 2 mm Eurofer stack tungsten-diffusion-limited

**Figure 3.** Hydrogen-isotope transport in the two stack materials from the cited data [4, 6]: (a) Arrhenius diffusivity and (b) Sieverts solubility for tungsten and Eurofer97 over the operating window. The tungsten solubility carries the larger activation energy, so the interface solubility partition $S_W/S_E \approx 2.4 \times 10^{-5}$ at 700 K makes the series stack tungsten-diffusion-limited (equation 10). Dotted lines mark the 600 and 800 K bounds.

Because the flux is diffusion-limited, it is temperature-independent and linear in the implanted flux (equation 9): the isothermal cases at 600, 700 and 800 K all return $6.00 \times 10^{14} \text{ T m}^{-2}\text{s}^{-1}$ ($= \varphi R_p/L_W$ exactly), and a tenfold rise in φ returns a tenfold flux (figure 5). This is the property that lets the fuel-cycle model of equation (2) carry first-wall permeation as a clean, load-scaled term: for a $\sim 100 \text{ m}^2$ first wall the loss is $\sim 8\text{--}10 \text{ g T}$ per full-power year into structure and coolant, a real but small holdup against a breeder tritium surplus measured in kilograms.

Table 2. FESTIM first-wall permeation and retained inventory for the 0.5 mm W + 2 mm Eurofer stack at implanted flux $\varphi = 1 \times 10^{20} \text{ m}^{-2} \text{ s}^{-1}$ (serial BR-L2-A25). The permeation flux is diffusion-limited and temperature-independent; only the retained inventory varies with temperature.

case	permeation flux ($\text{T m}^{-2}\text{s}^{-1}$)	retained inventory (T m^{-2})
operating gradient (800 $\rightarrow$ 600 K)	5.16×10^{14}	1.75×10^{18} ($\approx 8.8 \mu\text{g m}^{-2}$)
isothermal 600 K (worst case)	6.00×10^{14}	8.38×10^{19} ($\approx 0.42 \text{ mg m}^{-2}$)
isothermal 700 K	6.00×10^{14}	6.2×10^{18}
isothermal 800 K	6.00×10^{14}	9.5×10^{17}

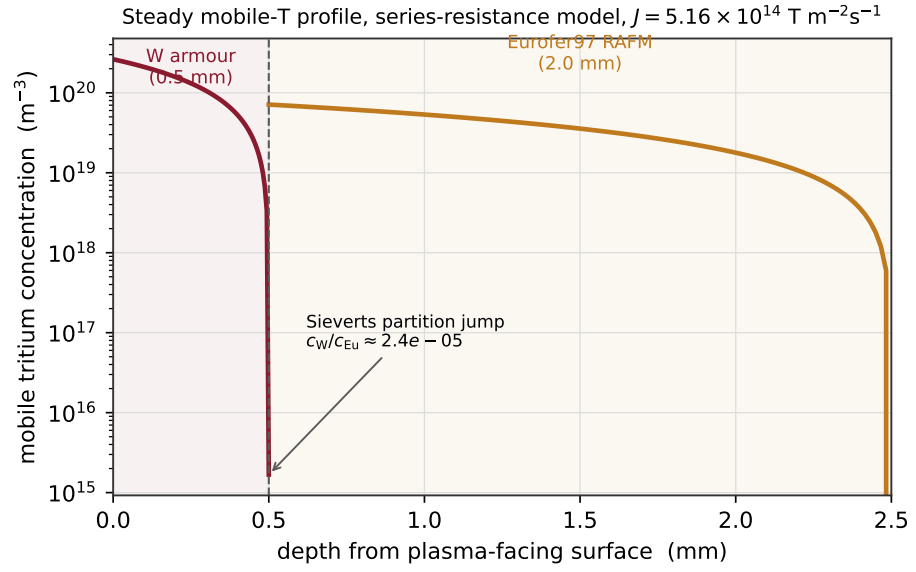


Figure 4. Computed steady mobile-tritium concentration across the bilayer from the two-layer series-resistance model at the operating flux $J = 5.16 \times 10^{14} \text{ T m}^{-2} \text{ s}^{-1}$. Almost the entire concentration drop is across the tungsten armour; the Sieverts chemical-potential continuity (equation 6) forces a $\sim 2 \times 10^{-5}$ concentration jump at the W/Eurofer interface, above which the steel carries the same flux at far higher concentration and negligible additional resistance.

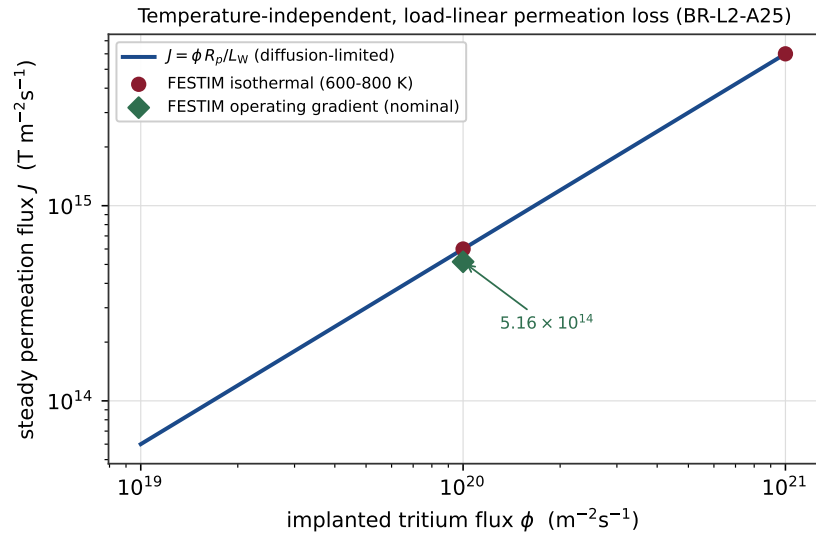


Figure 5. Steady permeation flux versus implanted flux. In the diffusion-limited implantation regime (equation 9) the flux is temperature-independent and linear in load: the finite-element isothermal points at $\varphi = 10^{20}$ and 10^{21} lie on $J = \varphi R_p/L_W$, and the operating-gradient case (5.16×10^{14}) sits just below the isothermal line. Serial BR-L2-A25.

4.3 Retained inventory and breakthrough time

The retained wall inventory (mobile plus trapped) is small and hot-wall-favoured (figure 6): $\approx 8.8 \mu\text{g m}^{-2}$ on the operating gradient, where the hot tungsten traps are mostly empty, rising to $\approx 0.42 \text{ mg m}^{-2}$ only in the cold 600 K isothermal bound, and falling to $9.5 \times 10^{17} \text{ T m}^{-2}$ at 800 K. The inventory is tungsten-dominated (the 1.00 eV trap). The transient is short: the analytic breakthrough time of equation (11), evaluated with the tungsten diffusivity, is 192 s at 600 K and 29 s at 800 K (figure 7), so permeation reaches steady within minutes of a burn start and holds no surprise transient. Loss flux and inventory are cleanly separated: the flux is set by geometry and load, the inventory by trap population and temperature.

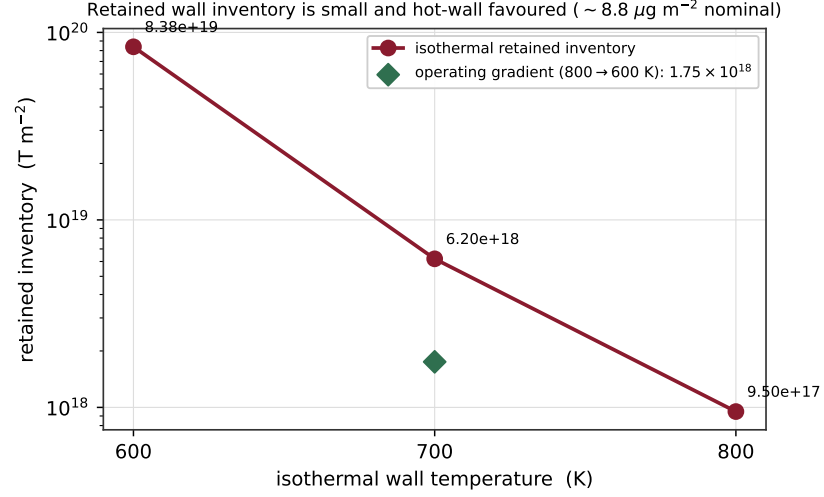


Figure 6. Retained tritium inventory versus wall temperature (serial BR-L2-A25). The inventory is hot-wall favoured — highest in the cold 600 K isothermal bound, lowest at 800 K — and the operating gradient case ($1.75 \times 10^{18} \text{ T m}^{-2} \approx 8.8 \mu\text{g m}^{-2}$) sits near the hot end. The diffusion-limited loss flux is unchanged across all four cases.

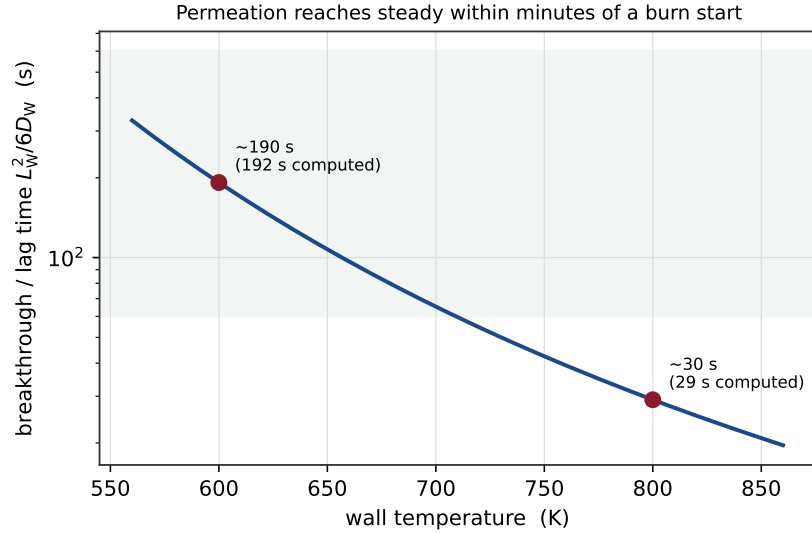


Figure 7. Breakthrough (time-lag) time $\tau = L_W^2/6D_W$ of the rate-controlling tungsten armour versus temperature (equation 11). The computed curve passes through the register anchors of $\sim 190 \text{ s}$ at 600 K and $\sim 30 \text{ s}$ at 800 K; permeation is at steady within minutes of a burn start.

4.4 Barrier ladder and closing the recovery loop

Even a sub-percent permeation loss is worth cutting when the coating is cheap in neutronic and thermal terms, and the standalone fuel-cycle target for Hyperion (BR-SX-17) is a permeation-reduction factor of order 2000. A PRF screen (equation 14) ranks the candidate barriers Al_2O_3 ($\text{PRF} \sim 10^3$) $>$ Er_2O_3 ($\sim 3 \times 10^2$) $>$ TiC $>$ TiN (figure 8). Alumina and erbia are

the standard fusion barrier oxides for good reason: a dense, adherent oxide is a low-permeability series layer that *multiplies* the tungsten-limited resistance already established, so the barrier and the diffusion limit act in series rather than in competition. No single oxide reaches PRF ~ 2000 alone, but stacked on the tungsten-limited baseline the combination clears the requirement. Recovery is closed by an yttrium hot-trap, whose hydride formation free energy of $\Delta G = -83$ to -97 kJ mol^{-1} makes it a hot sink that getters tritium from the lithium stream, and a downstream cold-trap that removes it at 92% efficiency. Retention (barrier) and recovery (getter train) together turn the first-wall tritium budget into a closed, accountable circuit — what little permeates is caught, and what is caught is measured — which is what the real-time tritium-accountancy twin [30] carries as a bounded holdup.

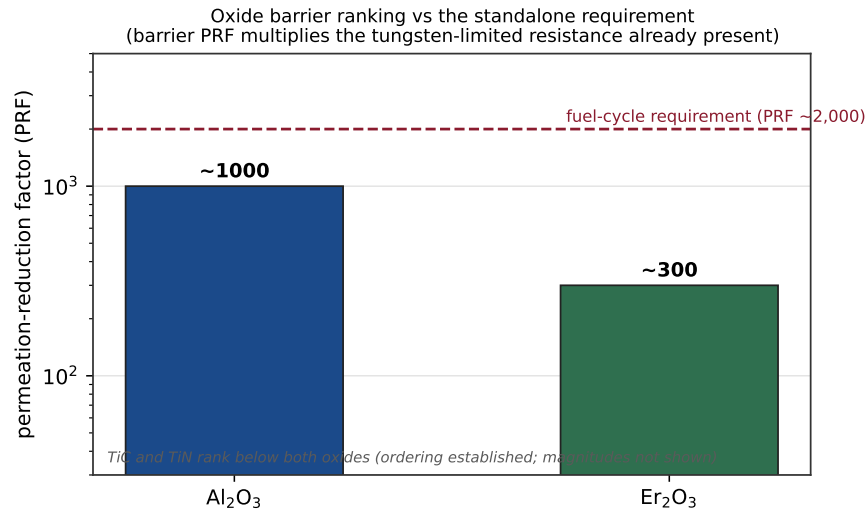


Figure 8. Measured permeation-reduction factors of the two leading oxide barriers (serial BR-SX-03) against the standalone fuel-cycle requirement of PRF ~ 2000 (serial BR-SX-17). Neither oxide reaches the requirement alone, but each multiplies the tungsten-limited resistance already present (equation 14); TiC and TiN rank below both oxides. Logarithmic axis.

4.5 Divertor integration: the exhaust co-benefit

The lithium choice pays a second dividend at the exhaust. A flowing-lithium film divertor handles a surface heat flux of 2.8, 4.9 and 6.7 MW m^{-2} at film velocities of 2, 5 and 10 ms^{-1} respectively (figure 9, HX-16), the limit set by the evaporation (surface-temperature) ceiling, which suits the detached-operation regime the breeder runs. A lithium vapour shield self-limits its surface near 869°C at 8 MW m^{-2} (BR-CX-25), adding a passive margin at the strike point. The same lithium loop that protects the fuel cycle at the first wall therefore also carries in-loop tritium recovery as a co-benefit and provides a self-healing exhaust surface — one material choice serving three functions — and it is the basis of the companion flowing-lithium vapour-box divertor with in-loop tritium extraction [31].

5 Discussion

5.1 The two decisions reinforce each other

The two first-wall decisions are not independent trades but a single coherent choice. Lithium's low tritium driving pressure (equation 12) is simultaneously what makes it the safe plasma-facing liquid and what makes it the liquid a barrier protects most easily, because the downstream flux it drives is already small. The tungsten-diffusion-limited stack (equation 10, figures 3–4) holds the baseline permeation loss under a percent of the implanted flux before any coating is applied. The alumina/erbia barrier plus yttrium getter train drives the residual loss down and routes it into a measured recovery stream. Tin is excluded on the one axis that matters, and lead–lithium keeps its proper role. The result is that first-wall permeation enters the self-sufficiency condition of equation (2) as a bounded, well-characterised term rather than an open risk.

5.2 Quantitative comparison with published benchmarks

The permeation number is consistent with, and more sharply bounded than, the fusion permeation literature. The tungsten-diffusion-limited character follows directly from the Frauenfelder solubility

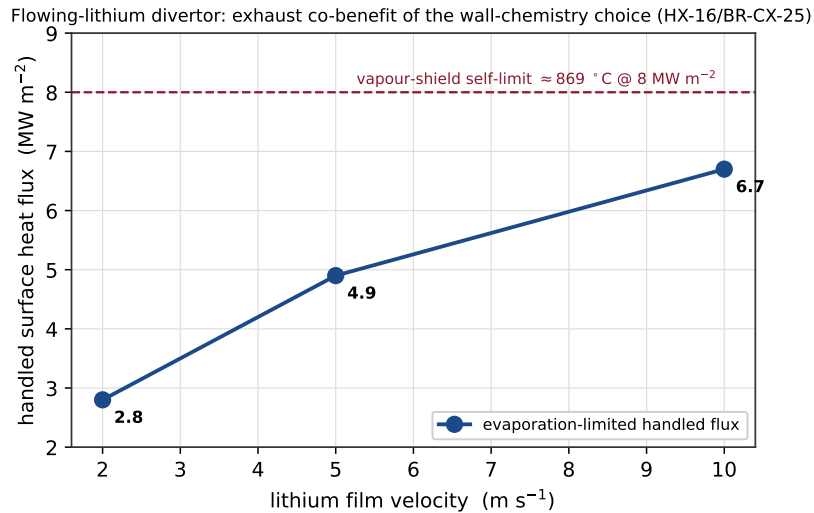


Figure 9. Evaporation-limited handled heat flux of the flowing-lithium film divertor versus film velocity (serial HX-16): 2.8, 4.9 and 6.7 MW m⁻² at 2, 5 and 10 m s⁻¹. The dashed line marks the vapour-shield self-limit (≈ 869 °C at 8 MW m⁻², BR-CX-25). The wall-chemistry choice that protects the fuel cycle also carries the exhaust.

being three-to-five decades below Eurofer’s over the operating window [4, 6], exactly the partition that measured permeation experiments on aluminised and oxide-coated steels invoke to explain their PRF [7, 16]. The diffusion-limited, temperature-independent, load-linear flux (equation 9, figure 5) is the standard Pick–Sonnenberg/Anderl implantation-driven result [10, 11], and recovering the analytic two-layer limit to seven significant figures places the finite-element solve firmly inside the verified TMAP permeation V&V family. The barrier ranking — $\text{Al}_2\text{O}_3 \sim 10^3$, $\text{Er}_2\text{O}_3 \sim 3 \times 10^2$ — sits within the 10^2 – 10^4 band reported for dense oxide permeation barriers [15, 17, 19, 18]. On the liquid side, the choice of lithium as a low-recycling, tritium-retaining plasma-facing material is consistent with the CDX-U/LTX/NSTX experience [20, 21] and with the analyses that reserve tin and tin–lithium for cases where lithium’s tritium inventory is judged unacceptable [23, 24]; here that inventory is precisely the property that protects the fuel cycle. Finally, treating first-wall permeation as a sub-percent, kilogram-margin loss term is what the global D–T self-sufficiency analyses require of a credible breeder [1, 3], and it feeds the Hyperion tritium breeding ladder [29] and accountancy twin [30] directly.

5.3 Consistency with the Hyperion design point

Nothing in the first-wall architecture conflicts with the frozen machine. The 0.5 mm tungsten armour and 2 mm Eurofer wall are the same stack the RAFM-steel qualification [32] and the first-wall neutron-damage lifetime study [33] carry, and the flowing-lithium exhaust is the same loop the divertor study [31] develops. The breeder operates at $P_{\text{fus}} = 85.04$ MW, $I_p = 9.66$ MA, $Q = 3.076$, $\delta = -0.30$, $B_0 = 8$ T behind a centrepost peak field of 16.84 T; the permeation loss is a platform-level fuel-cycle number, not a function of these plasma parameters beyond the implanted flux φ that sets its scale.

6 Limitations

The single open item is the barrier ranking. It is drawn here from established permeation-reduction data and oxide chemistry rather than computed from theory; the first-principles DFT hydrogen-solution energies for the Al_2O_3 and Er_2O_3 barriers, which would set their PRF from $E_{\text{sol}}(\text{H})$ rather than from measured factors, were not completed (the local MPI build was blocked) and are queued on the NVIDIA GPU HPC campaign. The permeation loss term itself is robust to this: being diffusion-limited it is unaffected by trap detail, and neutron-damage traps — omitted here — can only raise the retained inventory (by an estimated factor of 10–100), never the steady loss flux, which they merely delay. The retained-inventory numbers are therefore a clean-material lower bound, while the loss flux stands as computed. Two secondary conservatisms bracket the flux the safe way: the absence of a front recombination barrier places the permeation on the conservative-high side, and a thicker armour would lower it in proportion to $1/L_W$.

7 Conclusion

Lithium is the plasma-facing liquid for a compact spherical-tokamak breeder. It retains tritium some 6.7 orders of magnitude more strongly than tin — closing off the dominant fuel-cycle hazard through the low equilibrium driving pressure of equation (12) — and it hands the exhaust a $2.8\text{--}6.7\text{ MW m}^{-2}$ flowing-film surface as a bonus. Behind it, the real $0.5\text{ mm W} + 2\text{ mm Eurofer}$ stack leaks only $0.082\text{ g T m}^{-2}\text{ fpy}^{-1}$ — a computed, not assumed, number: tungsten-diffusion-limited, temperature-independent, load-linear, and matching the analytic two-layer limit to seven significant figures. An $\text{Al}_2\text{O}_3/\text{Er}_2\text{O}_3$ barrier and an yttrium getter train close the recovery loop into a measured circuit. Together these choices make the first-wall tritium budget a small, bounded and accountable term in the self-sufficiency condition $\Lambda = \text{TBR}(1 - \varepsilon) - 1 \geq 0$, and they do so with a single material choice that protects the fuel cycle, admits a barrier, and self-heals the exhaust at once.

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Reproducibility. The permeation and inventory numbers are frozen results (serial BR-L2-A25, FESTIM) in the Kronos Physics De-Risking Register [34], regenerable from the archived FESTIM driver, mesh and material-property table; the liquid-metal retention and barrier ranking are the M3 materials-chemistry gate (serial BR-SX-03), and the divertor thermofluids are HX-16/BR-CX-25. The figure-generation script deposited with this paper recomputes the transport coefficients, breakthrough times and solubility partition from the cited constants and reproduces the register anchors. This paper is archived at its DOI [doi:10.5281/zenodo.22645870](https://doi.org/10.5281/zenodo.22645870); official version: <https://www.kronosfusionenergy.com/publications/tritium-permeation-barrier-coatings-for-a-compact-spher.html> and its official page <https://www.kronosfusionenergy.com/publications/tritium-permeation-barrier-coatings-for-a-compact-spher.html>; the series register is [doi:10.5281/zenodo.22133057](https://doi.org/10.5281/zenodo.22133057).

Data availability

The FESTIM inputs and permeation/inventory outputs, the liquid-metal and barrier screen, and the figure-generation script for figures 2–9 are archived with the deposit at [doi:10.5281/zenodo.22645870](https://doi.org/10.5281/zenodo.22645870), which references the Kronos 2026 series register [doi:10.5281/zenodo.22133057](https://doi.org/10.5281/zenodo.22133057). Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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